

QUIZ

Objective

Implement a **Delta Live Tables (DLT) declarative pipeline** using Databricks Lakeflow to process material master data from multiple factories.

Dataset

- **File:** material_master_1k.csv
- **Format:** Pipe-delimited (|) CSV
- **Records:** 1,000 material master records
- **Source:** Multiple factory locations (PlantA, PlantB, PlantC, PlantD)

The screenshot displays the Databricks Lakeflow interface for a pipeline named 'master_material'. The pipeline is configured with a Python script that uses DLT to process material master data. The script includes column cleaning and selection logic, and a DLT table definition. The pipeline graph shows the flow from a bronze table to a silver table, which is then partitioned into two materialized views. The bottom table shows the execution results for these tables.

```
Python  
1  
.withColumn("material_id_clean", trim(upper(col("material_id"))))  
.withColumn("material_name_clean", trim(col("material_name")))  
.withColumn("plant_clean", trim(col("plant")))  
.withColumn("status_clean", trim(col("status")))  
  
.withColumn("silver_processed_timestamp", current_timestamp())  
  
.select(  
    col("material_id_clean").alias("material_id"),  
    col("material_name_clean").alias("material_name"),  
    col("category"),  
    col("plant_clean").alias("plant"),  
    col("status_clean").alias("status"),  
    col("unit_cost_clean").alias("unit_cost"),  
    col("supplier_name"),  
    col("country"),  
    col("last_updated_date").alias("last_updated"),  
    col("lead_time_days_int").alias("lead_time_days"),  
    col("safety_stock_int").alias("safety_stock"),  
    col("reorder_level_int").alias("reorder_level"),  
    col("ingestion_timestamp"),  
    col("silver_processed_timestamp")  
)  
  
@dlt.table(  
    ...  
)
```

Name	Catalog	Schema	Type	Duration	Output r...	Expectations	Dropped	Warnings	Failed	Incrementalization
bronze_material_master	workspace	default	Materialized view	8s	1K	Not defined	-	-	-	Full recompute
silver_material_master	workspace	default	Materialized view	7s	976	1 met 1 unmet	24	0	0	Full recompute
silver_material_master_quaran...	workspace	default	Materialized view	5s	24	Not defined	-	-	-	Full recompute

Tables 3

Performance 3

Filter by table

Type

Status

Name ↑	Catalog	Schema	Type	Duration	Output r...	Expectations	Dropped	Warnings	Failed	Incrementalization	
bronze_material_master	workspace	default	Materialized view	8s	1K	Not defined	-	-	-	Full recompute	
silver_material_master	workspace	default	Materialized view	7s	976	1 met 1 unmet	24	0	0	Full recompute	
silver_material_master_quaran...	workspace	default	Materialized view	5s	24	Not defined	-	-	-	Full recompute	

